

RHEL Enterprise Server Administration Project

Screenshots from my hands-on RHEL 10 lab

16 screenshots

github.com/MohamedOsama-16/rhel-enterprise-server

About this file

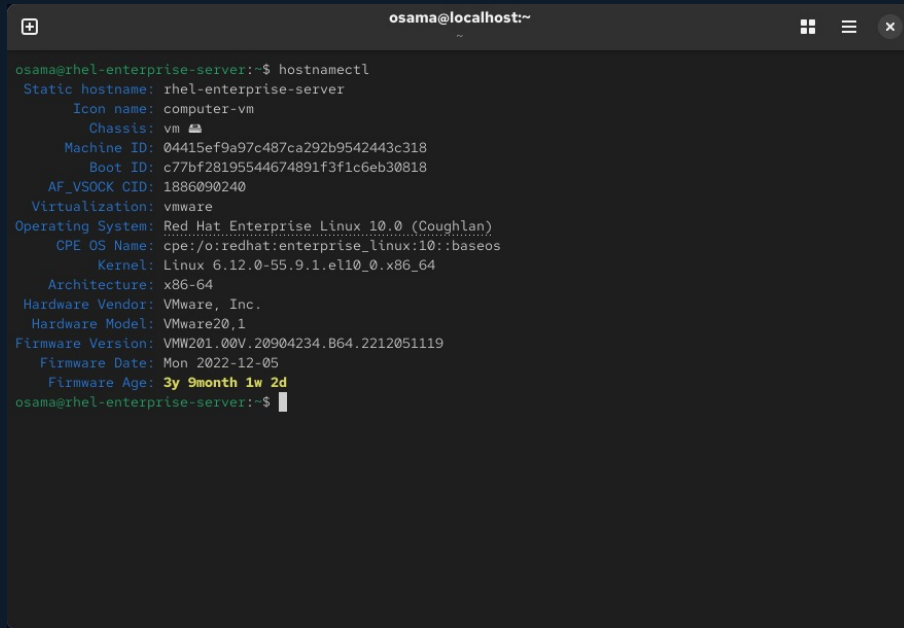
These are the screenshots I took while building and testing this project, in the order I did the work. The commands and notes are in the docs folder of the repo.

#	Section	Items
01	DEPLOYMENT & STORAGE	5
02	USERS, GROUPS & PERMISSIONS	3
03	NETWORKING & REMOTE ACCESS	3
04	SECURITY & SERVICES	2
05	AUTOMATION & BACKUP	2
06	FINAL VALIDATION	1

01 DEPLOYMENT & STORAGE

LVM, persistent mounts, and growing storage live.

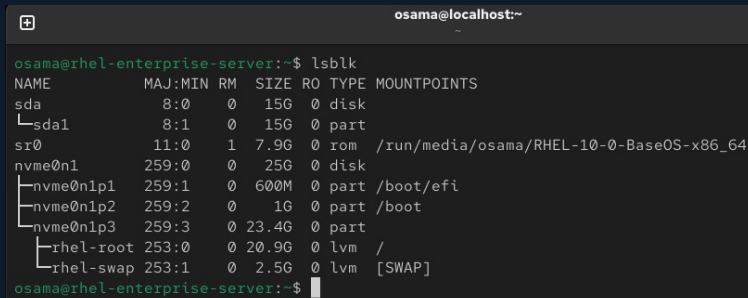
01 Initial Server Verification



```
osama@localhost:~  
osama@rhel-enterprise-server:~$ hostnamectl  
Static hostname: rhel-enterprise-server  
Icon name: computer-vm  
Chassis: vm  
Machine ID: 04415ef9a97c487ca292b9542443c318  
Boot ID: c77bf28195544674891f3f1c6eb30818  
AF_VSOCK CID: 1886090240  
Virtualization: vmware  
Operating System: Red Hat Enterprise Linux 10.0 (Coughlan)  
CPE OS Name: cpe:/o:redhat:enterprise_linux:10::baseos  
Kernel: Linux 6.12.0-55.9.1.el10_0.x86_64  
Architecture: x86-64  
Hardware Vendor: VMware, Inc.  
Hardware Model: VMware20,1  
Firmware Version: VMW201.00V.20904234.B64.2212051119  
Firmware Date: Mon 2022-12-05  
Firmware Age: 3y 9month 1w 2d  
osama@rhel-enterprise-server:~$
```

First check after install: RHEL 10.0 on VMware, hostname set.

02 Partition Created

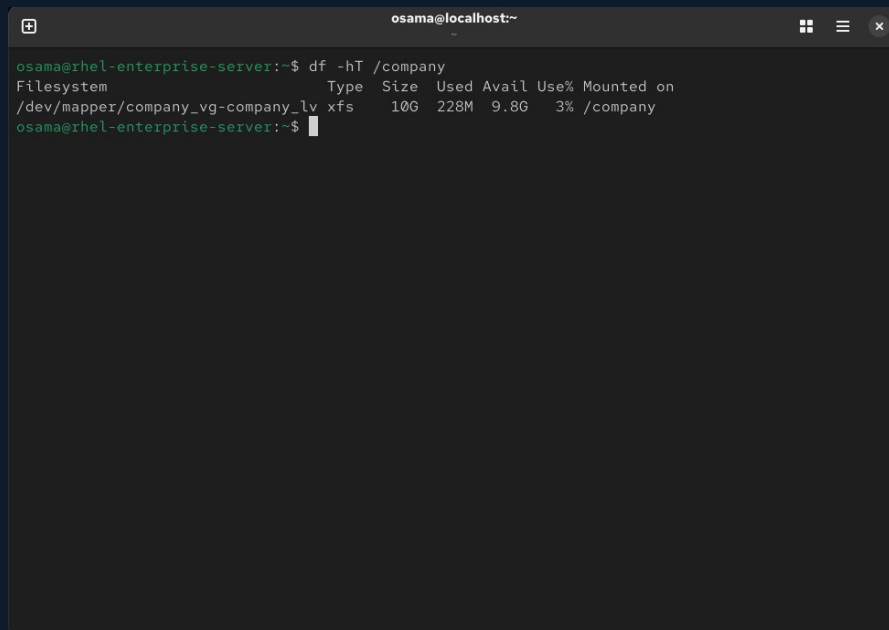


A terminal window titled 'osama@localhost:~' showing the output of the 'lsblk' command. The output is a tree-like structure of disk information. The 'sda' disk is shown with a single partition 'sda1' of 15G size, type 'part'. The 'nvme0n1' disk is shown with three partitions: 'nvme0n1p1' (600M, type 'part', mountpoint '/boot/efi'), 'nvme0n1p2' (1G, type 'part', mountpoint '/boot'), and 'nvme0n1p3' (23.4G, type 'part'). The 'rhel-root' and 'rhel-swap' are shown as LVM logical volumes of 20.9G and 2.5G respectively, both of type 'lvm'. The 'rhel-root' is mounted at '/', and 'rhel-swap' is marked as '[SWAP]'.

```
osama@rhel-enterprise-server:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
├─sda         8:0    0   15G  0 disk
│ └─sda1      8:1    0   15G  0 part
├─sr0        11:0    1    7.9G  0 rom  /run/media/osama/RHEL-10-0-BaseOS-x86_64
├─nvme0n1    259:0    0    25G  0 disk
│ ├─nvme0n1p1 259:1    0   600M  0 part /boot/efi
│ ├─nvme0n1p2 259:2    0    1G   0 part /boot
│ └─nvme0n1p3 259:3    0   23.4G  0 part
│   ├─rhel-root 253:0    0   20.9G  0 lvm  /
│   └─rhel-swap 253:1    0    2.5G  0 lvm  [SWAP]
osama@rhel-enterprise-server:~$
```

The 15GB data disk now has one partition (sda1), type Linux LVM.

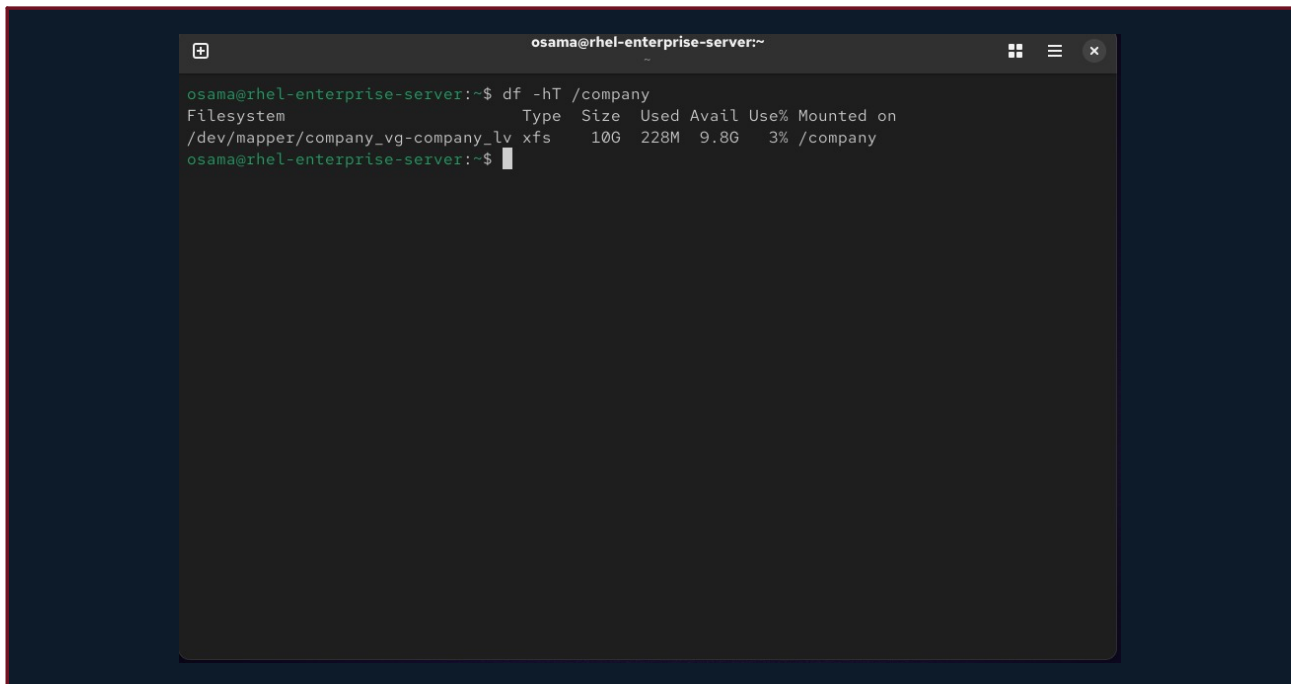
03 LVM Mount Success

A terminal window titled 'osama@localhost:~' showing the command 'df -hT /company' and its output. The output is a table with columns: Filesystem, Type, Size, Used, Avail, Use%, and Mounted on. The data row shows '/dev/mapper/company_vg-company_lv' as the filesystem, 'xfs' as the type, '10G' as size, '228M' as used space, '9.8G' as available space, '3%' as usage, and '/company' as the mount point.

```
osama@rhel-enterprise-server:~$ df -hT /company
Filesystem                                Type      Size  Used Avail Use% Mounted on
/dev/mapper/company_vg-company_lv         xfs       10G   228M   9.8G   3% /company
osama@rhel-enterprise-server:~$
```

LVM is ready and /company is mounted as XFS, 10GB.

04 Persistent Mount Verified

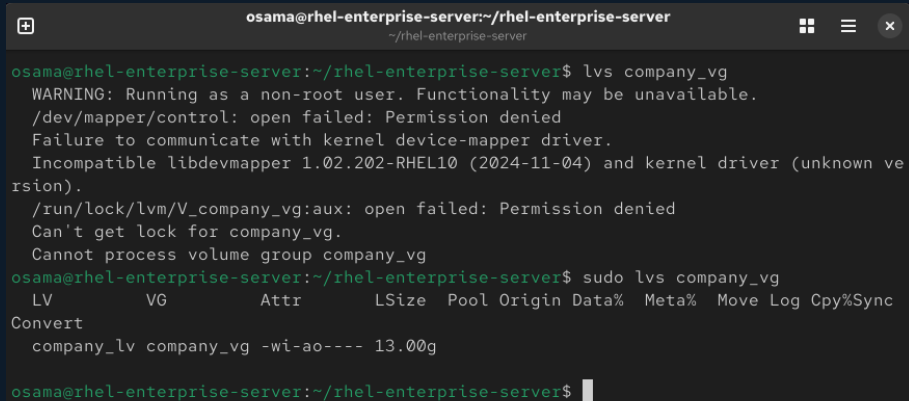
A terminal window titled 'osama@rhel-enterprise-server:~' showing the command 'df -hT /company' and its output. The output is a table with columns: Filesystem, Type, Size, Used, Avail, Use%, and Mounted on. The data row shows '/dev/mapper/company_vg-company_lv' as the filesystem, 'xfs' as the type, '10G' as size, '228M' as used space, '9.8G' as available space, '3%' as usage, and '/company' as the mount point.

```
osama@rhel-enterprise-server:~$ df -hT /company
Filesystem                                Type      Size  Used Avail Use% Mounted on
/dev/mapper/company_vg-company_lv         xfs       10G   228M  9.8G   3% /company
osama@rhel-enterprise-server:~$
```

Same check after a full reboot: /company came back on its own because of the fstab entry.

05

Storage Expansion Verified

A terminal window titled 'osama@rhel-enterprise-server:~/rhel-enterprise-server' with a dark background. It shows the execution of 'lvs company_vg' which fails with permission errors. Then 'sudo lvs company_vg' is run, showing a table of LVM components. The 'company_lv' is listed with a size of 13.00g.

```
osama@rhel-enterprise-server:~/rhel-enterprise-server$ lvs company_vg
WARNING: Running as a non-root user. Functionality may be unavailable.
/dev/mapper/control: open failed: Permission denied
Failure to communicate with kernel device-mapper driver.
Incompatible libdevmapper 1.02.202-RHEL10 (2024-11-04) and kernel driver (unknown ve
rsion).
/run/lock/lvm/V_company_vg:aux: open failed: Permission denied
Can't get lock for company_vg.
Cannot process volume group company_vg
osama@rhel-enterprise-server:~/rhel-enterprise-server$ sudo lvs company_vg
LV          VG          Attr          LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync
Convert
company_lv  company_vg  -wi-ao---- 13.00g

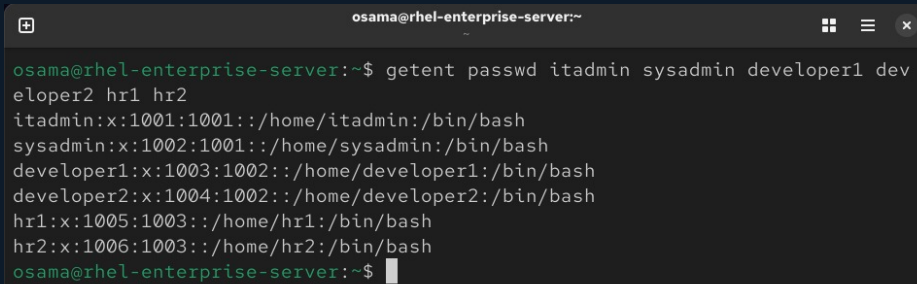
osama@rhel-enterprise-server:~/rhel-enterprise-server$
```

Grew company_lv from 10GB to 13GB while the server was running.

02 USERS, GROUPS & PERMISSIONS

Departments, permissions, SGID and ACL.

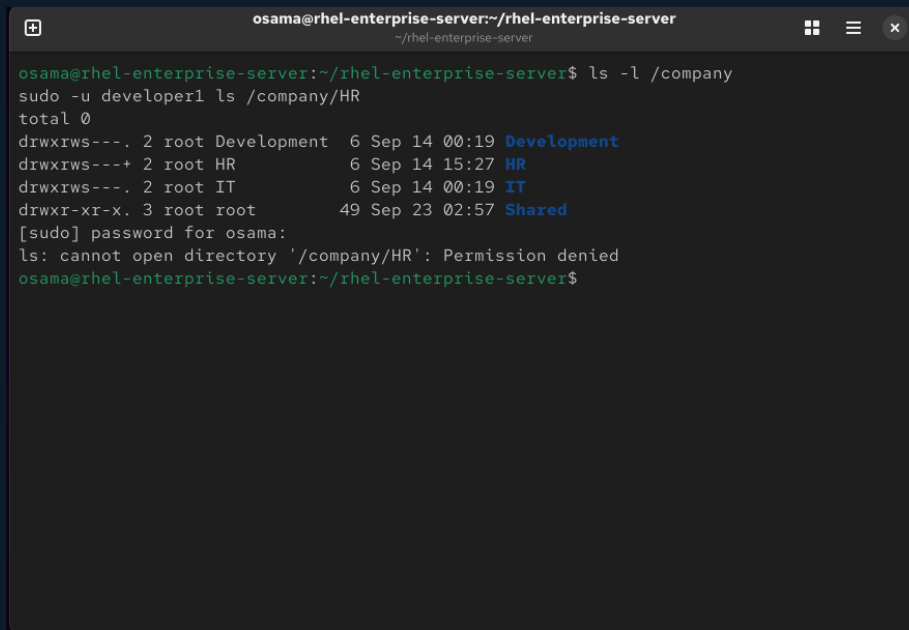
06 Users & Groups Created



```
osama@rhel-enterprise-server:~$ getent passwd itadmin sysadmin developer1 developer2 hr1 hr2
itadmin:x:1001:1001::/home/itadmin:/bin/bash
sysadmin:x:1002:1001::/home/sysadmin:/bin/bash
developer1:x:1003:1002::/home/developer1:/bin/bash
developer2:x:1004:1002::/home/developer2:/bin/bash
hr1:x:1005:1003::/home/hr1:/bin/bash
hr2:x:1006:1003::/home/hr2:/bin/bash
osama@rhel-enterprise-server:~$
```

The 6 test users and their groups (IT, Development, HR).

07 Permissions Tested & Denied



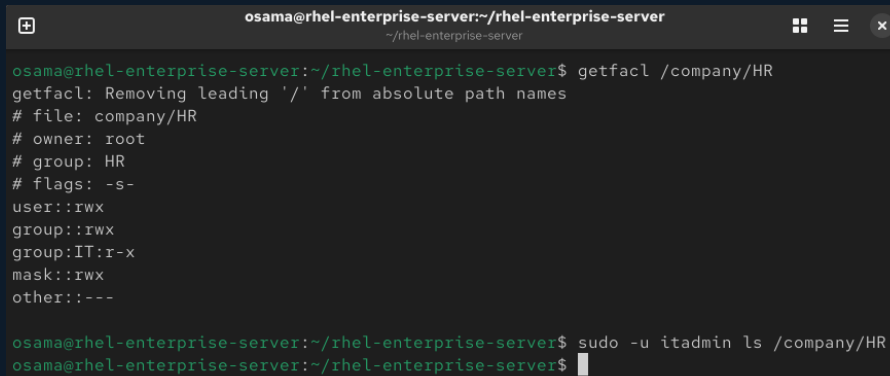
```
osama@rhel-enterprise-server:~/rhel-enterprise-server
~ /rhel-enterprise-server

osama@rhel-enterprise-server:~/rhel-enterprise-server$ ls -l /company
sudo -u developer1 ls /company/HR
total 0
drwxrws---. 2 root Development  6 Sep 14 00:19 Development
drwxrws---+ 2 root HR           6 Sep 14 15:27 HR
drwxrws---. 2 root IT           6 Sep 14 00:19 IT
drwxr-xr-x. 3 root root         49 Sep 23 02:57 Shared
[sudo] password for osama:
ls: cannot open directory '/company/HR': Permission denied
osama@rhel-enterprise-server:~/rhel-enterprise-server$
```

developer1 tries to open /company/HR and gets Permission denied, as expected.

08

ACL Scenario Verified



```
osama@rhel-enterprise-server:~/rhel-enterprise-server
osama@rhel-enterprise-server:~/rhel-enterprise-server$ getfacl /company/HR
getfacl: Removing leading '/' from absolute path names
# file: company/HR
# owner: root
# group: HR
# flags: -s-
user::rwx
group::rwx
group:IT:r-x
mask::rwx
other:---

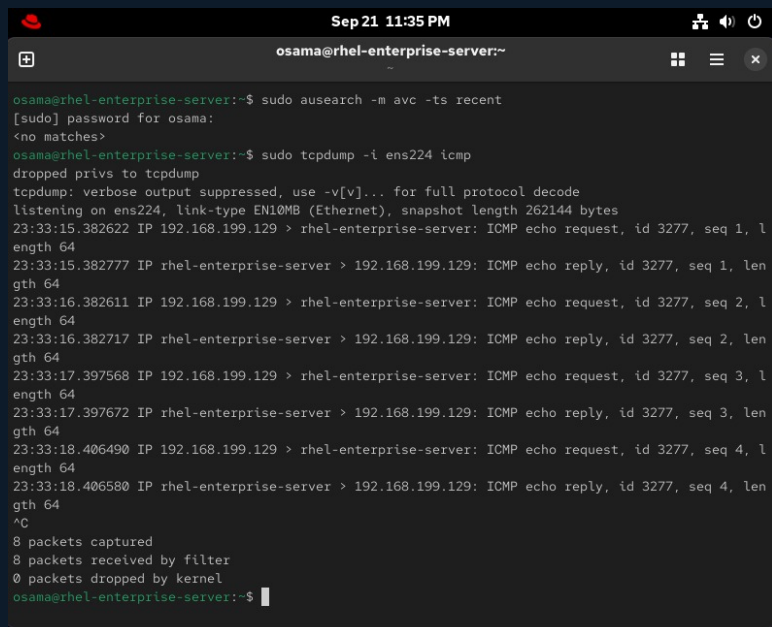
osama@rhel-enterprise-server:~/rhel-enterprise-server$ sudo -u itadmin ls /company/HR
osama@rhel-enterprise-server:~/rhel-enterprise-server$
```

ACL gives IT read-only on HR. itadmin can list it, developer1 still can't.

03 NETWORKING & REMOTE ACCESS

Second network, a real ping problem, and SSH.

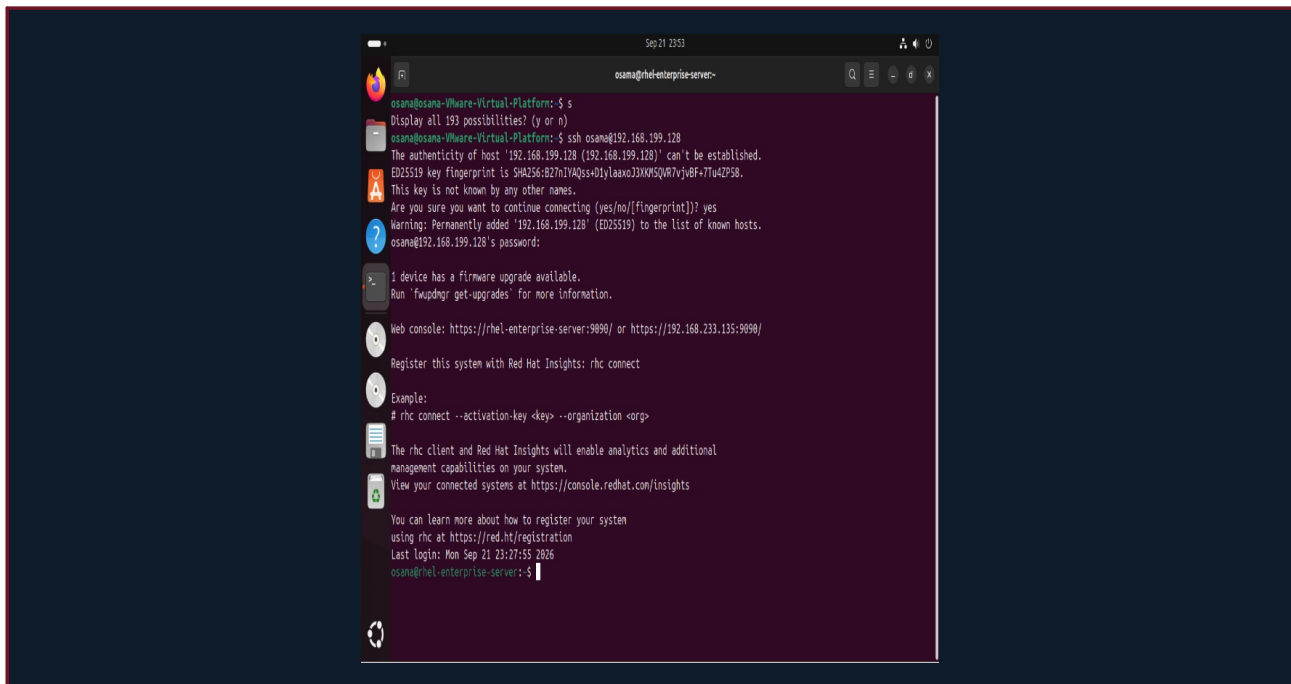
09 Network Connectivity Verified



```
osama@rhel-enterprise-server:~$ sudo ausearch -m avc -ts recent
[sudo] password for osama:
<no matches>
osama@rhel-enterprise-server:~$ sudo tcpdump -i ens224 icmp
dropped privs to tcpdump
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens224, link-type EN10MB (Ethernet), snapshot length 262144 bytes
23:33:15.382622 IP 192.168.199.129 > rhel-enterprise-server: ICMP echo request, id 3277, seq 1, length 64
23:33:15.382777 IP rhel-enterprise-server > 192.168.199.129: ICMP echo reply, id 3277, seq 1, length 64
23:33:16.382611 IP 192.168.199.129 > rhel-enterprise-server: ICMP echo request, id 3277, seq 2, length 64
23:33:16.382717 IP rhel-enterprise-server > 192.168.199.129: ICMP echo reply, id 3277, seq 2, length 64
23:33:17.397568 IP 192.168.199.129 > rhel-enterprise-server: ICMP echo request, id 3277, seq 3, length 64
23:33:17.397672 IP rhel-enterprise-server > 192.168.199.129: ICMP echo reply, id 3277, seq 3, length 64
23:33:18.406490 IP 192.168.199.129 > rhel-enterprise-server: ICMP echo request, id 3277, seq 4, length 64
23:33:18.406580 IP rhel-enterprise-server > 192.168.199.129: ICMP echo reply, id 3277, seq 4, length 64
^C
8 packets captured
8 packets received by filter
0 packets dropped by kernel
osama@rhel-enterprise-server:~$
```

Ping from Ubuntu to RHEL kept failing, while the other direction worked. I ruled out firewalld, ufw and SELinux first, then ran tcpdump on RHEL to see whether the packets were arriving at all. The problem turned out to be a typo in my own ping command (I forgot the number after -c). Once I fixed it, tcpdump showed the requests and replies going through.

10 SSH First Connection Success



The screenshot shows a terminal window titled 'osana@rhel-enterprise-server-'. The user is in a shell on a system named 'osana@osana-Virtual-Platform'. They run the command 'ssh osana@192.168.199.128'. The terminal displays the following output:

```
osana@osana-Virtual-Platform:~$ ssh osana@192.168.199.128
The authenticity of host '192.168.199.128 (192.168.199.128)' can't be established.
ED25519 key fingerprint is SHA256:B27n1VAQss+Dilyaaxo13XKMSQWR7vj8F+7Tu4ZPS8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.199.128' (ED25519) to the list of known hosts.
osana@192.168.199.128's password:
osana@rhel-enterprise-server:~$
```

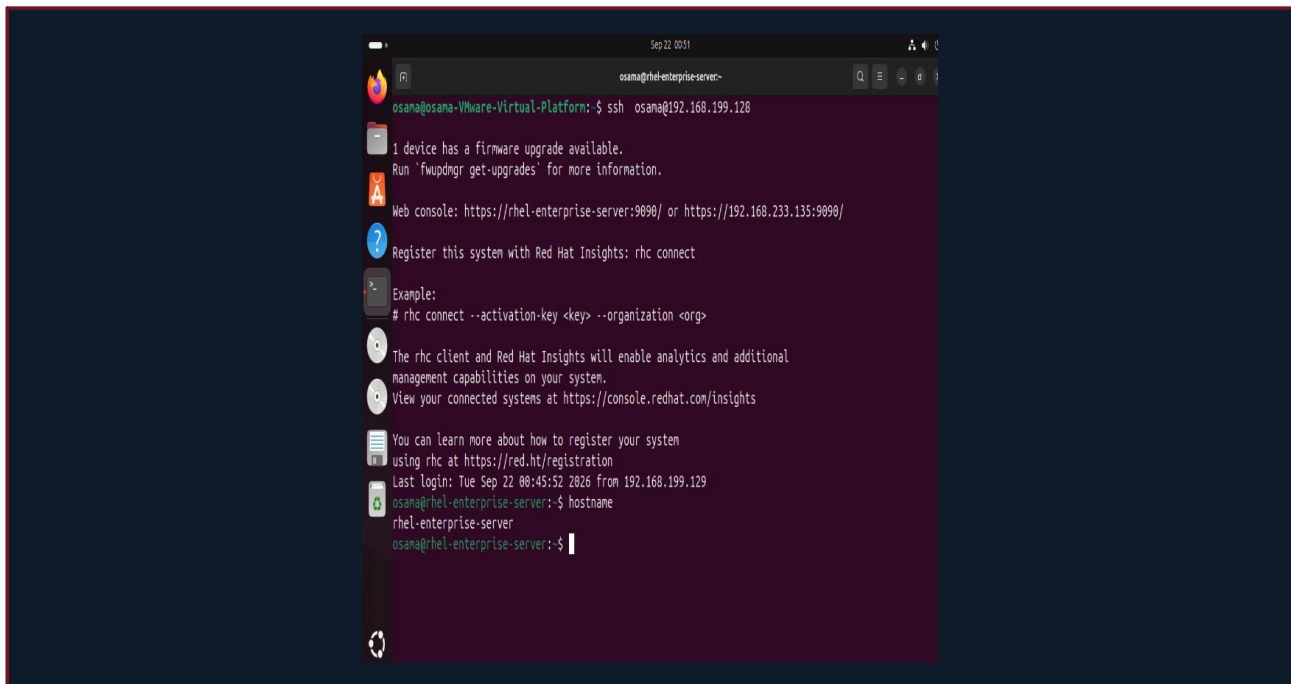
Below the terminal window, there are several system notifications:

- A message: "1 device has a firmware upgrade available. Run 'fwupdgr get-upgrades' for more information."
- A link to the web console: "Web console: <https://rhel-enterprise-server:9090/> or <https://192.168.233.135:9090/>"
- A prompt to register the system with Red Hat Insights: "Register this system with Red Hat Insights: `rhc connect`"
- An example command: "Example: `# rhc connect --activation-key <key> --organization <org>`"
- A message: "The rhc client and Red Hat Insights will enable analytics and additional management capabilities on your system."
- A link to view connected systems: "View your connected systems at <https://console.redhat.com/insights>"
- A message: "You can learn more about how to register your system using rhc at <https://red.ht/registration>"
- A login message: "Last login: Mon Sep 21 23:27:55 2026"

First SSH from Ubuntu to RHEL, with a password.

11

SSH Key Authentication Success

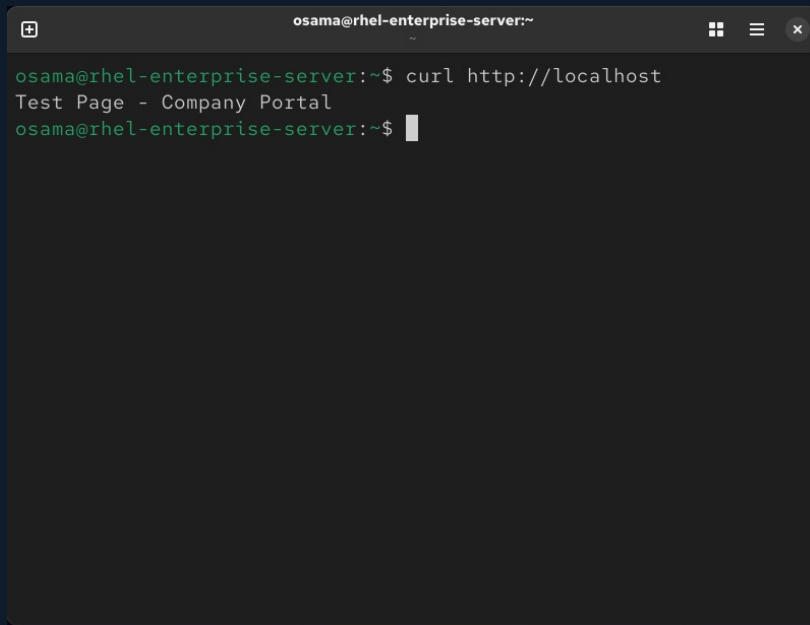


Same SSH login, now with a key and no password prompt.

04 SECURITY & SERVICES

SELinux and Apache.

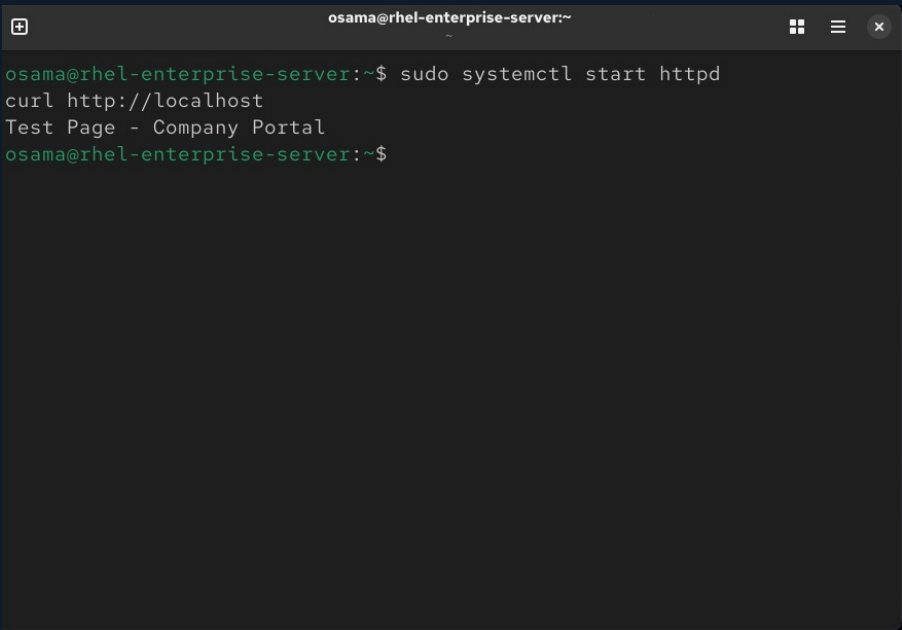
12 SELinux Troubleshooting Resolved

A terminal window titled 'osama@rhel-enterprise-server:~' with standard window controls. It shows a user running the command 'curl http://localhost' which returns 'Test Page - Company Portal'. The prompt returns to 'osama@rhel-enterprise-server:~\$' with a cursor.

```
osama@rhel-enterprise-server:~$ curl http://localhost
Test Page - Company Portal
osama@rhel-enterprise-server:~$
```

After I pointed Apache to a new web folder, I got 403 Forbidden. The file permissions looked fine, so I checked the SELinux label and it was wrong (unlabeled_t). I fixed it, but I still got 403. I switched SELinux to permissive for a minute just to test, and it was still 403, so SELinux wasn't the real cause (I put it back to enforcing right after). The actual problem was in Apache: any folder outside /var/www needs a Directory block with "Require all granted". I added one and the page loaded.

13 Apache Down Troubleshooting

A terminal window titled 'osama@rhel-enterprise-server:~' with standard window controls. It shows the execution of 'sudo systemctl start httpd' followed by 'curl http://localhost'. The output of the curl command is 'Test Page - Company Portal'.

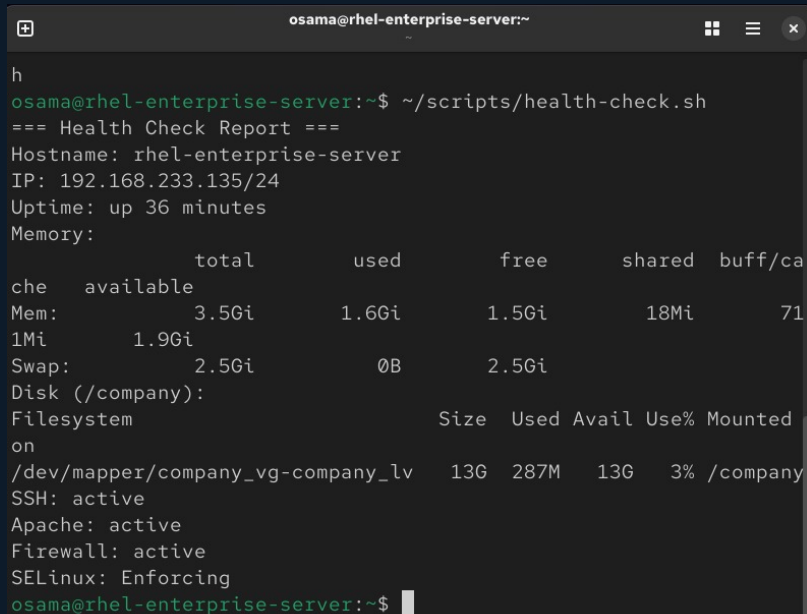
```
osama@rhel-enterprise-server:~$ sudo systemctl start httpd
curl http://localhost
Test Page - Company Portal
osama@rhel-enterprise-server:~$
```

Apache running again and answering curl after I stopped it on purpose.

05 AUTOMATION & BACKUP

The health check script and a real restore test.

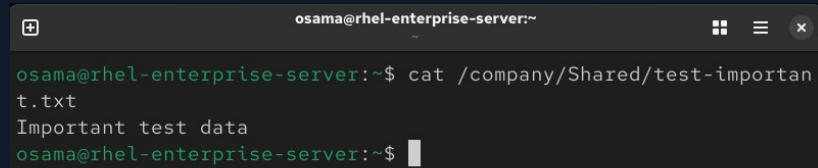
14 Health Check Script Success



```
osama@rhel-enterprise-server:~  
h  
osama@rhel-enterprise-server:~$ ~/scripts/health-check.sh  
=== Health Check Report ===  
Hostname: rhel-enterprise-server  
IP: 192.168.233.135/24  
Uptime: up 36 minutes  
Memory:  
      total      used      free      shared  buff/ca  
che  available  
Mem:   3.5Gi      1.6Gi      1.5Gi      18Mi      71  
1Mi    1.9Gi  
Swap:   2.5Gi          0B      2.5Gi  
Disk (/company):  
Filesystem              Size  Used Avail Use% Mounted  
on  
/dev/mapper/company_vg-company_lv 13G  287M  13G   3% /company  
SSH: active  
Apache: active  
Firewall: active  
SELinux: Enforcing  
osama@rhel-enterprise-server:~$
```

Output of my health-check.sh script.

15 Backup & Restore Verified

A terminal window titled 'osama@rhel-enterprise-server:~' with standard window controls. The prompt is 'osama@rhel-enterprise-server:~\$'. The command 'cat /company/Shared/test-important.txt' has been entered and executed. The output of the command is 'Important test data'. The prompt is now 'osama@rhel-enterprise-server:~\$' with a cursor.

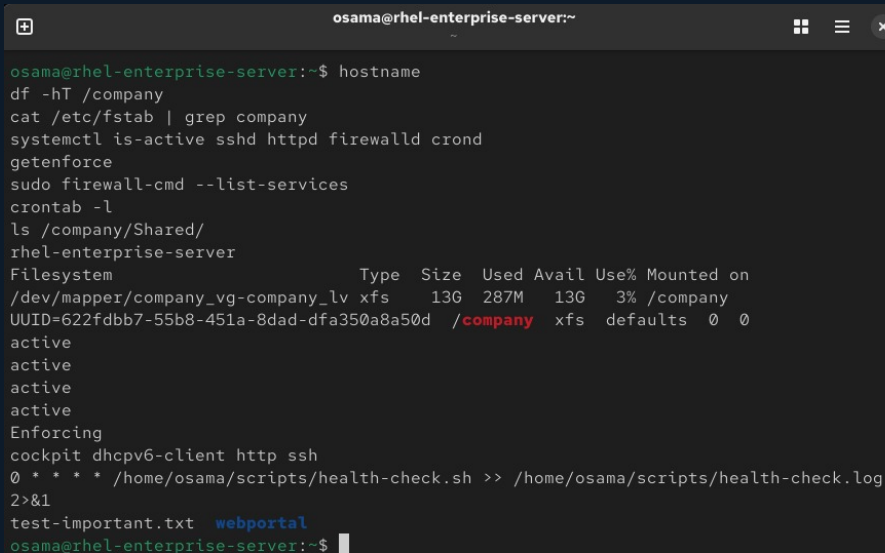
```
osama@rhel-enterprise-server:~$ cat /company/Shared/test-important.txt
Important test data
osama@rhel-enterprise-server:~$
```

The restored test file with its content intact. I deleted it first, then restored it from the tar backup.

06 FINAL VALIDATION

Everything after a full reboot.

16 Final Validation Post-Reboot



```
osama@rhel-enterprise-server:~$ hostname
osama@rhel-enterprise-server:~$ df -hT /company
osama@rhel-enterprise-server:~$ cat /etc/fstab | grep company
osama@rhel-enterprise-server:~$ systemctl is-active sshd httpd firewalld crond
osama@rhel-enterprise-server:~$ getenforce
osama@rhel-enterprise-server:~$ sudo firewall-cmd --list-services
osama@rhel-enterprise-server:~$ crontab -l
osama@rhel-enterprise-server:~$ ls /company/Shared/
osama@rhel-enterprise-server:~$ rhel-enterprise-server
osama@rhel-enterprise-server:~$
Filesystem                                Type      Size      Used Avail Use% Mounted on
/dev/mapper/company_vg-company_lv        xfs        13G      287M    13G   3% /company
UUID=622fdbb7-55b8-451a-8dad-dfa350a8a50d /company  xfs        defaults 0 0
active
active
active
active
Enforcing
cockpit dhcpv6-client http ssh
0 * * * * /home/osama/scripts/health-check.sh >> /home/osama/scripts/health-check.log
2>&1
test-important.txt webportal
osama@rhel-enterprise-server:~$
```

After a full reboot: mount, services, firewall, SELinux and cron all came back by themselves.